Improving Opinion Estimation from Tweets by Collaborative Filtering *

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Abstract

In this paper, we consider the task of inferring opinions of individuals about specific items/concepts based on their tweet data. If a person has tweeted about an item, then from its sentiment analysis it is possible to infer his/her opinion about the item. Such an opinion estimate, however, is at best noisy given the limited accuracy of sentiment classifiers. Moreover, in cases where a person has *not* tweeted about an item, this direct method fails to provide an opinion estimate. In this paper, we explore the use of collaborative filtering to filter out noise in the opinion estimates as well as to extrapolate them to unseen items. We consider a corpus of about 700,000 tweets from 572 users and focus on 166 nouns occurring frequently in these tweets. Using tweet sentiment analysis, we estimate about 3.9% of the user-noun opinion matrix. The opinions are binary and for our dataset, we find that about 72% known entries are +1. We apply collaborative filtering to this sparsely known user-noun matrix. Specifically, using OptSpace, a matrix completion algorithm, we obtain opinion prediction accuracy of 81%.

1 Introduction

Twitter provides a rich platform to share short, real-time messages. It is used by several million people everyday for sharing personal updates, co-ordinating campaigns, and so on. Many businesses also use it to directly communicate with their customers and manage their brands. The massive conversation on Twitter contains several interesting signals: flu outbreaks can be tracked from tweets ((Paul and Dredze 2011)), situational awareness in disaster zones may be possible from tweets ((Verma and others 2011)), and aggregate sentiment about items/people/concepts can also be inferred from tweets (Go, Bhayani, and Huang 2009), (Pak and Paroubek 2010). This paper deals with opinion estimation from tweets. However, unlike previous works (for example, (Go, Bhayani, and Huang 2009), (Pak and Paroubek 2010)), we are not interested in finding the aggregate sentiment over a population; rather we are interested in studying whether it is possible to *infer the sentiment of a specific individual about a specific item/concept*. This is a challenging task for two reasons.

- Sentiment analysis of individual tweets is imperfect and at best yields noisy estimates. Therefore, if we wish to estimate opinion of a specific individual about a specific item, then we need to find some other information that helps us average out the noise in sentiment analysis of tweets.
- If an individual has not tweeted about an item, then there is no direct evidence of the sentiment. In other words, we have to figure out a way to extrapolate the opinion of that individual from items he/she has tweeted about to unseen items.

The right tool to achieve these two ends is *collaborative filtering* (see, for example, (Koren 2008; Keshavan, Montanari, and Oh 2010; Aditya, Dabeer, and Dey 2011; Barman and Dabeer 2010)), which pools together data from different users. Our aim is to explore whether collaborative filtering used in conjunction with existing sentiment analysis techniques like (Go, Bhayani, and Huang 2009) yields good estimates of individuals' opinions.

To attain our goal, we picked an initial list of users from the Twitter graph using a random walk, and then shortlisted only those who showed significant activity. We collected about 700,000 tweets from these users and identified the most frequent proper nouns, using a mix of parts of speech tagger and some manual shortlisting. We applied likelihood metrics similar to those in sentiment analysis and then thresholded them to form a binary, i.e., positive-negative, *user-noun sentiment matrix*. The matrix obtained is sparse - only about 3.9% of the entries are known. Of these, we hide 30% entries and then predict them using the remaining 70% by employing collaborative filtering. We get 81% accuracy in prediction with a standard deviation of about 1% over the random hiding realizations. The details of data collection and processing to form the user-noun matrix are given in Section 2. The results are reported in 3. We conclude in Section 4.

2 Obtaining the User-Noun Matrix

User Selection for Obtaining Tweet Corpus

The idea of collaborative filtering hinges on the expectation that the users and nouns under consideration have related be-

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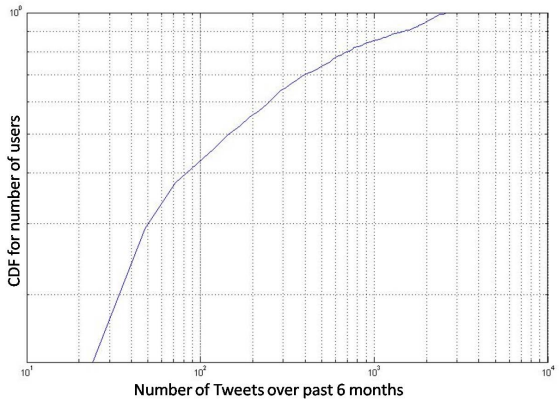

Figure 1: Histogram of the number of tweets in last six months by 2475 users obtained from the random walk.

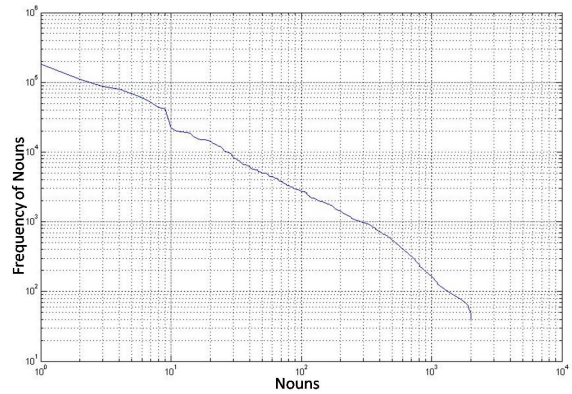

Figure 2: The frequency of nouns from the tweet corpus in decreasing order.

havior, which leads to a low-rank user-noun matrix. Thus, if we pick a random sample of Twitter users, then the users may not have many commonalities, and collaborative filtering is not expected to be effective. At the other extreme, if we pick Twitter users forming a strong clique, then we expect their opinions to be correlated and, therefore, we expect collaborative filtering to work well. In our data collection, we wanted to ensure that we avoid both these extremes. Hence to find Twitter users of interest, we picked a user with a high number of followers and performed a random walk on the Twitter graph starting from this node. The random walk mixes rapidly in the component of the graph that is well connected to a root node. Thus, if we stop the random walk after a finite number of steps, then the nodes visited by the random walk can be expected to be close to each other in the Twitter graph. The random walk also ensures that we have a random sample and we do not pick a clique. In other words, the random walk allows us to pick nodes close to each other in the Twitter graph, but which are not strongly connected. We identified 2475 users in this fashion. In Figure 1, we plot the cumulative distribution function of the number of tweets by these users in 6 months. We see that a large number of them are not very active. We, therefore, considered only those users who have 180 tweets or more in the last 6 months (an average of 1 or more per day). This yielded a short list of 572 users. A tweet corpus of size about 0.7 million tweets from these users was downloaded and used for further analysis.

Noun Tagging and Short Listing

Ideally, we would like to find opinions about items or concepts commonly tweeted by the chosen users. Since we chose a random sample of Twitter users, we do not know a good list of items for the chosen population *a priori*. Since our aim is to give a proof of concept, instead of dealing with items, we decided to deal with nouns. That is, we identify popular nouns arising in the tweet corpus and estimate the sentiment that the users have about these nouns. Towards this end, we apply the Stanford Parts of Speech Tagger to our tweet corpus and identify nouns in the tweet corpus. Figure 2 shows the frequency-rank plot (frequency of the nouns arranged in decreasing order). We picked 300 most frequent nouns whose frequency is about 1000 or more. We found that many of the nouns short listed are common nouns such as 'man', 'child', 'cars', etc. These were removed manually to finally obtain a list of 166 nouns. In Table 1, we list the 20 most frequent nouns chosen in this manner.

Twitter	iPhone	Facebook
BlackBerry	foursquare	Google
Android	iPad	UberSocial
Mac	Silver	Ford
Pluggio	GM	Twidroyd

Table 1: List of 15 most frequent nouns in our list of 166 nouns.

Forming Opinion Scores

To estimate opinions of users about nouns, we use a simple scheme similar to Naive Bayes sentiment classification (Go, Bhayani, and Huang 2009). We use the training set of 40,000 tweets from (Go, Bhayani, and Huang 2009), which has been tagged positive or negative using emoticons. The two classes are equal in size. This training set is independent of our tweet corpus and is only used to learn a classifier. We use unigrams (excluding the emoticons) as features. For each unigram w, let $N_+(w)$ ($N_-(w)$) be the number of times it occurs in the class of positive (negative) tweets from the training set. The likelihood of w is defined to be $\ell(w) = \log(N_+(w)/N_-(w))$. If the likelihood is positive, then the unigram is more frequent in positive tweets than negative tweets and vice-versa. For any unigram not appearing in the training set, we set $\ell(w) = 0$indicating that we can say nothing about the sentiment from such unigrams.

Given a tweet by user i containing noun j, we first parse it into unigrams $w_1, ..., w_n$ and assign the score $S(i,j) = \sum_{k=1}^{n} \ell(w_k)$. User i may tweet about noun j several times.

In this case, we add the scores for all these tweets to form the final score $S(i,j)$. If $S(i,j)$ is very large (small), then we are confident that the sentiment of user i about noun j is positive (negative). The values close to zero indicate low confidence. Hence we use the following thresholding:

$$X(i,j) = \text{sign}(S(i,j)), |S(i,j)| > 3,$$
$$= 0, \text{ otherwise.}$$

This is our user-noun sentiment matrix denoted by X. We note that a zero indicates that we do not know the opinion - either we have no tweets from user i about noun j, or even if we have data, the confidence is low and we do not assign a sentiment. The threshold value of 3 leads to about 3.9% non-zero entries in the matrix X. A higher value of the threshold leads to fewer entries making collaborative filtering hard, while a lower threshold lets more noise into the matrix X.

3 Collaborative Filtering

In the previous section, we generated the user-noun sentiment matrix X. Only 3.9% of the entries of this matrix are known. However, if we expect the opinions of users to be related, then we expect the complete matrix X to be approximated by a low-rank matrix. This is a commonly used assumption in collaborative filtering of movie rating matrices (Koren 2008; Keshavan, Montanari, and Oh 2010; Aditya, Dabeer, and Dey 2011). Due to the low-rank structure, we expect to be able to make predictions based only on a few samples. We next explore whether we can predict entries in the matrix X. Towards this end, we randomly hide about 30% non-zero entries from X and try to predict these hidden entries based on the remaining 70% entries. The aim is to predict the sign of the hidden entries. We count the number of errors we make and refer to the fraction of errors as classification error (CE). All CE values reported below are averaged over 10 random realizations of the hiding set.

Remark: We note that in the computation of CE, we treat the hidden entries in X as the absolute ground truth. While in general these entries are also noisy estimates of opinions, we have more confidence in them for the following reason.

- a) Each nonzero entry of X is usually derived from several tweets.
- b) By thresholding the values $S(i,j)$ with a dead-zone around 0, we only retain those opinions with high absolute value of the likelihood, that is, high confidence.

A Simple Benchmark

We find that about 72% percent of the non-zero entries in X are +1. Thus if we always guess the hidden entries as +1, our failure rate is about 28%. For collaborative filtering to be effective, it should do better than this simple estimator.

OptSpace

There are several methods for matrix completion. We used a method called OptSpace, which consists of some pruning and singular value decomposition techniques (see (Keshavan, Montanari, and Oh 2010) for details). The algorithm has a parameter r, the rank of the low-rank approximation we are seeking. From Figure 3, we see that the best performance is obtained for rank equal to 2 and the corresponding CE is 19%, which is better than the simple benchmark above. The standard deviation over the random choice of hidden entries is also small - about 0.01. This result shows that collaborative filtering in conjunction with the sentiment analysis of tweets can allow us to extrapolate opinions of users to unseen items.

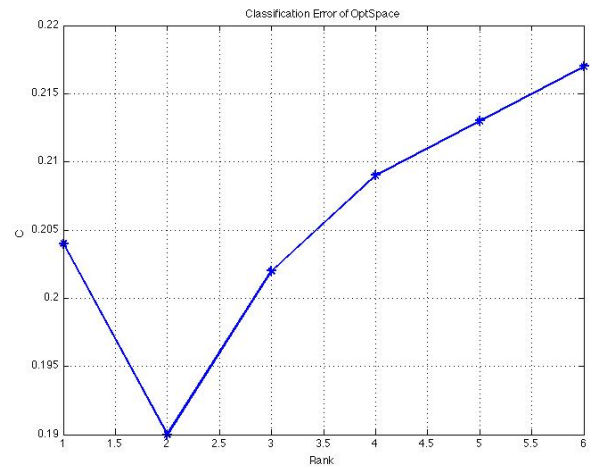

Figure 3: The minimum classification error of OptSpace is obtained at rank parameter equal to 2.

A closer look at the CE shows that the 19% error rate consists of two error rates: for hidden entries that are +1, the error rate is 13%, while the error rate for hidden entries that are -1 is 34%. Thus, negative tweets seems to be harder to predict.

4 Conclusion

In this paper our aim was to explore whether collaborative filtering in conjunction with existing sentiment analysis techniques for tweets can be used for estimating opinions of individuals about specific items/concepts. We collected tweet data from users selected via a random walk on the Twitter graph and extracted frequent nouns using parts of speech tagger and shortlisted them manually. The user-noun rating matrix was generated using techniques similar to the classifier in (Go, Bhayani, and Huang 2009), and OptSpace method was applied for predicting missing entries. The net result is that we got a prediction accuracy of about 81% with a standard deviation of about 1%. This is better than the naive estimate, which always declares the sentiment to be positive and has accuracy of 72%. These results give hope that we can infer opinions individuals may have about items/concepts even though they may have never tweeted about them directly. However, much more needs to be done. First, we need to collect and test on other data samples from Twitter. Second, several algorithmic variations - different sentiment classifiers, different threshold param-

eters, different matrix completion algorithms - need to be tried and tested.

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